

SCHOOL ID APP — WIREFRAME SPEC (v1.0 BUILD DOCUMENT)

GLOBAL APP STRUCTURE

User Roles

- Student (passive, mostly scanned)
 - Staff (primary operator)
 - Admin (configuration + data control)
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CORE NAVIGATION (STAFF APP)

Bottom Tab Navigation:

1. **Scan (Home)**
 2. **Live Dashboard**
 3. **Activities**
 4. **Students**
 5. **More / Admin (role-based)**
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SCREEN 1 — SCAN (HOME SCREEN)

Purpose:

Primary operational screen. Fast scanning + instant action.

Layout:

Top Section:

- School Name
 - Current Time (live)
 - Staff Name / Role
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Center (Primary Action Area):

- **Camera View (Full Width)**
 - QR / Barcode scanning active by default
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Bottom Overlay (Action Panel — appears after scan):

When student is scanned → show card:

Student Card:

- Name
 - Photo
 - Grade/Class
 - Current Status (color-coded):
 -  On Campus
 -  In Class
 -  At Event
 -  Absent
 -  Checked Out
-

Action Buttons (Context-Aware):

-  Check-In (Morning)
 -  Check-Out (End of Day / Early Leave)
 -  Mark Class Attendance
 -  Mark Event Attendance
 -  Add Merit / Demerit
 -  Register Activity (Club / Detention)
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Special Case UI:

If inconsistent:

- ⚠️ “Student not checked in but marked for class”
 - ⚠️ “Already checked out”
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Data Sources:

- Student profile
 - Last scan event
 - Daily state engine
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SCREEN 2 — LIVE DASHBOARD (MISSION CONTROL)

Purpose:

Real-time situational awareness (EXCEPTION-FIRST)

Layout:

Top KPI Bar:

- Total Students
 - Present Today
 - Absent
 - Checked Out
 - Unaccounted ⚠️ (highlighted red)
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Critical Alerts Panel (TOP PRIORITY):

- Students missing from class
- Students missing from assembly
- Students unaccounted

- Late arrivals
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Live Activity Feed:

Chronological stream:

- “John checked in (07:32)”
 - “Sarah marked absent in Grade 8B”
 - “Michael checked out (Sick)”
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Status Distribution Panel:

- Pie / segmented bars:
 - On Campus
 - In Class
 - At Event
 - Checked Out
-

Quick Filters:

- Grade
 - Class
 - Event
 - Status
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Drill-down List:

Clickable list:

- Opens student-level detail
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Data Sources:

- Scan events (gate, class, event)
- State engine reconciliation

- Activity logs
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SCREEN 3 — ACTIVITIES / EVENTS



Purpose:

Manage structured attendance beyond normal classes



Layout:

Top Tabs:

- Assemblies
 - Events
 - Clubs
 - Detention
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Event List:

Each item:

- Event Name
 - Time
 - Responsible Staff
 - Status (Active / Completed)
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Event Detail Screen:

- Event Info
- Expected Students
- Checked-In Count
- Missing Students ⚠

Action:

- “Start Scanning”
→ Opens camera with event context
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Data Sources:

- Admin-defined activities
 - Student lists per activity
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SCREEN 4 — STUDENTS



Purpose:

Lookup + audit individual students



Layout:

Search Bar:

- Name / ID / QR scan shortcut
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Student List:

- Name
 - Grade
 - Status (color-coded)
-

Student Profile Screen:

Header:

- Photo
 - Name
 - Grade/Class
-

Status Panel:

- Current Status
 - Last Seen (timestamp + location)
-

Timeline (VERY IMPORTANT):

- Check-in
 - Class attendance
 - Events
 - Check-out
-

Behavior Panel:

- Merits / Demerits history
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Actions:

- Manual override (admin only)
 - Assign behavior
 - Force check-in/out
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Data Sources:

- Student record
- Event logs
- Behavior logs

SCREEN 5 — ADMIN / CONFIGURATION

Purpose:

System control + customization

Layout:

Sections:

1. Data Management

- Upload Excel (students/staff)
 - Manual add student
 - Edit student records
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2. Activity Configuration

- Create:
 - Assemblies
 - Events
 - Clubs
 - Detention types
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3. Behavior System

- Define:
 - Merits
 - Demerits

- Point values
 - Categories
-

4. Roles & Permissions

- Staff roles
 - Admin access control
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5. System Settings

- School hours
 - Attendance rules
 - Late thresholds
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Data Sources:

- Admin database
 - Configuration tables
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CORE SYSTEM LOGIC (CRITICAL FOR DEV)



STUDENT STATE ENGINE

At any moment, each student is:

- NOT ARRIVED
- ON CAMPUS
- IN CLASS
- AT EVENT
- CHECKED OUT
- ⚠️ UNACCOUNTED



STATE TRANSITIONS (Driven by scan events)

Examples:

- Gate scan → ON CAMPUS
 - Class scan → IN CLASS
 - Event scan → AT EVENT
 - Checkout scan → CHECKED OUT
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EXCEPTION ENGINE

Triggers:

- In class but not checked in
 - Missing from event
 - Not in class during period
 - Disappears from system flow
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UX PRINCIPLES (MANDATORY)

1. **Scan-first design (not menu-first)**
 2. **Exception-first dashboards**
 3. **Minimal taps (max 2 actions after scan)**
 4. **Color-coded states everywhere**
 5. **Real-time updates (Socket-based)**
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BACKEND EXPECTATIONS (FOR REPLIT)

- API:

- /scan
 - /students
 - /events
 - /dashboard/live
 - /behavior
 - Database:
 - students
 - scan_events
 - activities
 - behavior_logs
 - users
 - Real-time:
 - WebSocket / Socket.IO for live dashboard
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WHAT THIS GIVES YOUR DEVELOPER

This wireframe spec enables:

- ✓ Immediate UI scaffolding
 - ✓ Clear backend structure
 - ✓ Defined state machine
 - ✓ Defined workflows
 - ✓ No ambiguity on purpose
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